Arjun Mamidala

arjunmamidala88@gmail.com | +1 (919) 428-3795 | LinkedIn | GitHub | Portfolio

Education

North Carolina State University | GPA: 3.7/4.0 — Raleigh, NC
B.S. in Computer Science — *Expected December 2026*

- **Coursework:** Data Structures & Algorithms, Operating Systems, Software Engineering, Intro to Artificial Intelligence, ML & Data Analysis, Neural Networks, C & Software Tools, NLP
- **Honors:** National Cyber Scholar (2024) · Best Use of Gemini API – HackUNCP (2026)

Technical Skills/Certifications

Certifications: ISC2 Certified in Cybersecurity, AWS Certified Cloud Practitioner, CompTIA A+
Languages: Java, Python, C, JavaScript/TypeScript, SQL, Go, Bash
Frameworks: React, Node.js, Next.js, Flask, FastAPI, Spring Boot, PyTorch, LangGraph, Pandas, scikit-learn
AI/Developer Tools: Git, Docker, Jenkins, Pinecone, Valgrind, Postman, MySQL Workbench, GitHub Copilot, Claude Code, Cursor, Windsurf

Experience

Software Engineer Intern — June 2026 – August 2026
Fidelity Investments — *Durham, NC*

- Built a full-stack Batch Governance Assessor with an HTML/JavaScript frontend and Python FastAPI backend, streamlining intake submission and handling 50+ form fields across 9 domains with attachment storage to Confluence.
- Engineered 15+ RESTful APIs for Confluence integration and self-serve intake assessments, enabling iterative revisions that eliminated manual admin review cycles and reduced review turnaround time by 80–90%.
- Refactored sequential validation requests into parallel asynchronous REST calls, reducing validation latency and improving responsiveness across the assessment workflow.

Undergraduate Researcher — Jan 2026 – Present
AI-Assisted Learning Lab — *Raleigh, NC*

- Developed behavior-informed Deep Knowledge Tracing models for programming practice, using 195K submissions from 919 CS1 students across two semesters (CodeWorkout) and 116K next-attempt prediction targets.
- Engineered and compared two behavioral representations, validated with student-disjoint cross-validation and bootstrap tests: seven Attempt-Level Signals per submission and SPAM-mined Sequential Behavior Patterns. Attempt-Level Signals raised cross-semester AUC from 82.14% to 83.51%, and the patterns helped most on first submissions for a problem.
- Co-author of a paper from the AI-Assisted Learning Lab, submitted to SIGCSE TS 2027.

Software Engineer Intern — May 2025 – July 2025
Cisco — *Durham, NC*

- Built a Node.js Webex chatbot with a command-based query system that retrieves security requirements across build environments, serving 50+ engineers across 4 teams and reducing manual lookup time from 5–10 minutes to seconds.
- Integrated Cisco's internal AI API to support natural language queries on top of the existing command system, enabling engineers to retrieve security requirements conversationally instead of memorizing command syntax.
- Collaborated with engineers across 4 teams to gather requirements and iteratively refine response formatting for 50+ supported commands, increasing chatbot adoption after deployment.

Cyber Intern — May 2024 – Aug 2024
Align Technology — *Morrisville, NC*

- Conducted SAST on 2 applications, identifying and facilitating the fix of 15+ code vulnerabilities.
- Proposed and planned a Jeopardy-style CTF challenge to boost security awareness for 500+ employees.

Projects

Distributed Fault-Tolerant Key-Value Store | *Go, Raft, net/rpc, Docker*

- Implemented Raft-based distributed KV store in Go with leader election, log replication, snapshot compaction, and crash recovery across a 5-node cluster.
- Built a chaos testing framework with fault injection (leader crashes, partitions, cascading failures) validating post-fault convergence; benchmarked 5 concurrent clients at 234 ops/sec puts, 285 ops/sec gets, and p99 $<$ 26ms.
- Replaced simulated in-process RPC with real TCP networking (Go net/rpc), enabling each Raft node to run as an isolated process; containerized 5-node cluster with Docker Compose for one-command deployment.

Structural Defect Detection System | *Python, PyTorch, YOLOv8m-seg, React, FastAPI, Docker, PostgreSQL, LangGraph*

- Fine-tuned YOLOv8m-seg for pixel-level crack instance segmentation on 617 manually polygon-annotated drone images, achieving 0.54 mask mAP@0.5.
- Engineered confidence calibration via Platt scaling (ECE: 0.022) and FHWA-standard severity scoring using crack skeletonization with automated low-confidence flagging for manual review.
- Deployed end-to-end inspection pipeline with FastAPI backend, React + Tailwind frontend, and PostgreSQL inspection history storing per-image detections, severity scores, and confidence values across sessions.
- Implemented a LangGraph-based reporting workflow using GPT-4o with severity-aware conditional routing, transforming calibrated crack detections into inspection narratives and severity-prioritized maintenance recommendations for field inspectors.